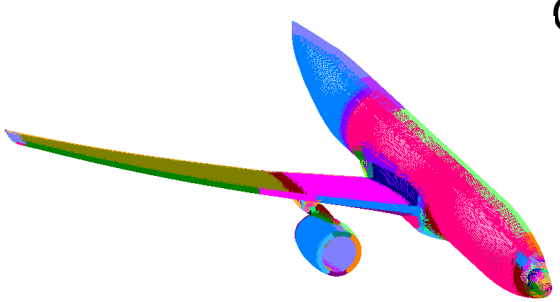
OVERVIEW OF CGT 3.0 AND A TUTORIAL ON THE EPOGS AUTOMATIC OVERSET MESH GENERATION TOOLSET

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Computational Aerosciences Branch

NASA Ames Research Center





17th Symposium on Overset Composite Grids and Solution Technology,
State College, PA, September 21-24, 2026

OUTLINE

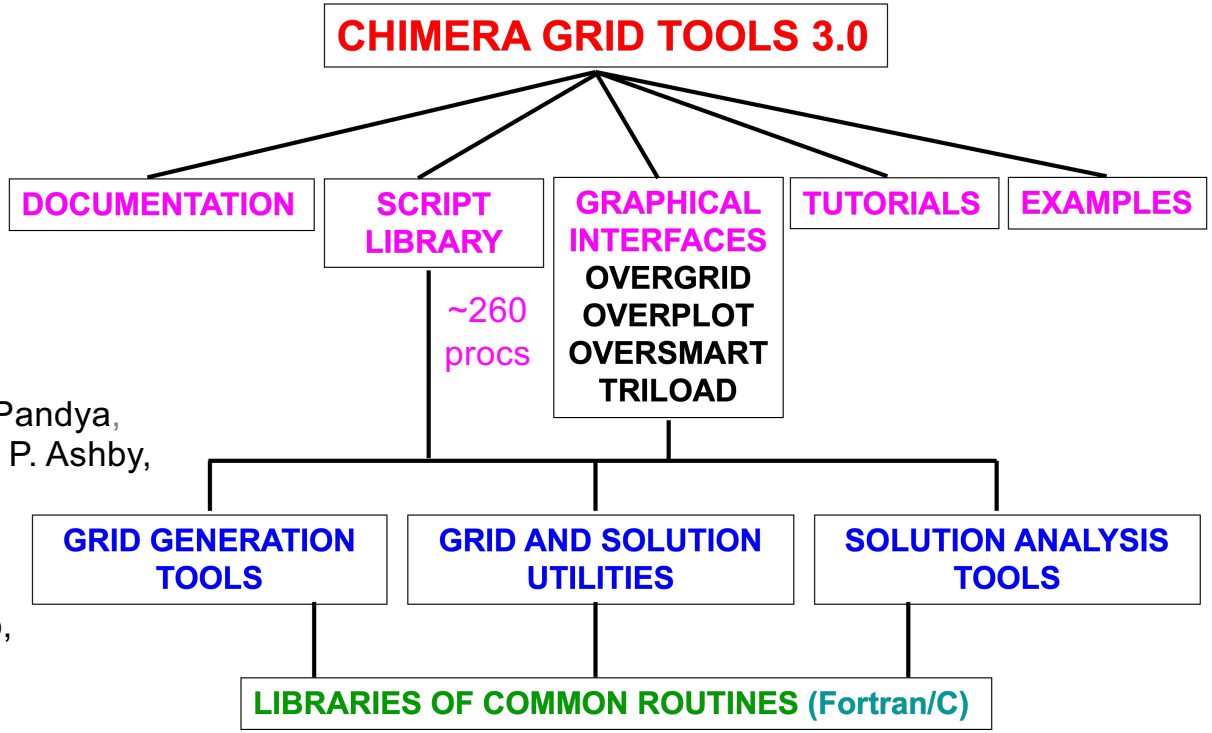
- Overview of Chimera Grid Tools 3.0
- Geometry preparation
- Overview of auto-meshing toolset EPOGS
- EPOGS demo

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Chimera Grid Tools (CGT) Overview

What is CGT

- A collection of software tools
  for pre- and post-processing
  of overset grid CFD computations



Authors

Currently active: William Chan, Shishir Pandya, Andrew Chuen, Chase P. Ashby, Nicholas J. Peters, Darby Vicker

Previous versions: Stuart Rogers
James Jensen, Henry Lee, David Kao, Pieter Buning, Robert Meakin, David Boger, Steve Nash

Availability

NASA Software Catalog (https://software.nasa.gov/software/ARC-16025-1C)
- U.S. citizens/permanent residents working under U.S. organization in the U.S.

Lines ~ 820k
Tools = 79
EPOGS ~ 30%
OVERGRID ~20%

CGT 3.0 Installation, Documentation, Tutorials

- Software requirements
  - Fortran, C, C++ compilers
    - GNU (gfortran, gcc, g++) 8, 9, 10, 11, 12, 13, 14, 15
    - Intel (ifort, icc, icpc) 2023, 2024
    - Intel (ifx, icx, icpx) 2024
  - OpenGL, X11, XQuartz 2.8.6 (Mac OS26+), Tcl/Tk up to 8.6.17 (~~9.0+~~): OVERGRID
  - Python, swig, gnuplot : OVERSMART
  - Tcl wish, xmgrace or gnuplot : OVERPLOT
  - Engineering Sketch Pad (ESP) needed for EPOGS (analytic geometry starting point)

- Operating systems - Linux, Mac (supported), Windows/WSL (unsupported)

- Installation instruction - chimera3.0/doc/INSTALLATION.html

- Documentation - chimera3.0/doc/CGTmain.html -> *.html

- Tutorials
  - chimera3.0/tutorials/* (OVERGRID, Script Library, EPOGS)
  - https://www.nas.nasa.gov/pubs/ams/2014/{05-13-14,05-20-14,05-29-14}.html

Compilation, Executables, Main Tools

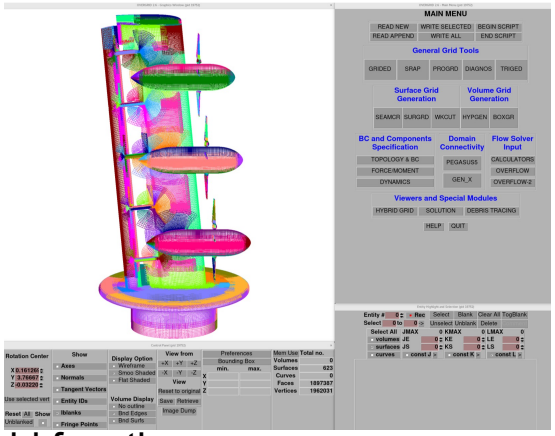
- Run configure script to generate Makefiles

  configure --help (get list of options)
  configure --enable-omp (recommended, with setenv OMP_NUM_THREADS #p)
  configure --debug (activate debug mode)

- Executables
  - single & double precision set

- FFIO Library automates file attributes determination and i/o

- overConvert used for file conversion



- OVERGRID – GUI for grid & solution viewing, access to most grid functions

- Script Library – Tcl procs with 1-line command to access most grid functions

- EPOGS – toolset for overset mesh generation automation

New in Version 3.0

- Last public release 2.2 was in 2018
- Main updates
  - EPOGS
    - EGADS2SRF (Electronic Geometry Aircraft Design to Surfaces)
    - POGS (Pre-Processor for Overset Grid Simulations)
  - FFIO Library Improvements
  - BladeGen
- Many smaller updates listed in chimera3.0/doc/cgt3.0.html

FFIO LIBRARAY

- Fortran library for handling I/O of files in the overset grid simulation workflow
  - Structured grid (Plot3d)
  - Surface triangulation (Cart3d)
  - Connectivity data (XINTOUT, INTOUT, CINTOUT)
  - X-ray map
  - Solution (Plot3d, OVERFLOW)
  - Function (Plot3d)

- Automatic detection and I/O for
  - single precis / double precis / ascii
  - little / big endian
  - with / without iblanks
  - multiple / single zone

- Automatic handling of large single records
  - Subrecords
  - Binary (no record markers)

- Grid splitting
  - Needed for OVERFLOW
  - Not needed for LAVA

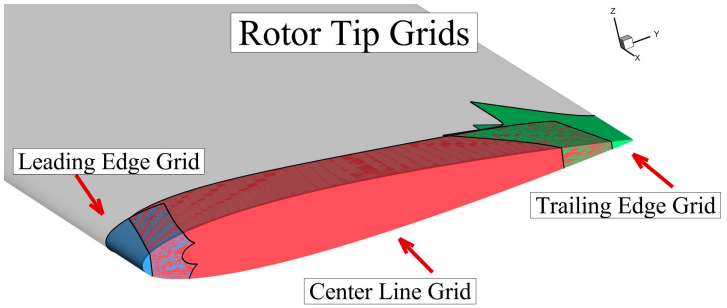
- All commonly used tools in CGT 3.0 are now FFIO enabled

BLADEGEN TOOL

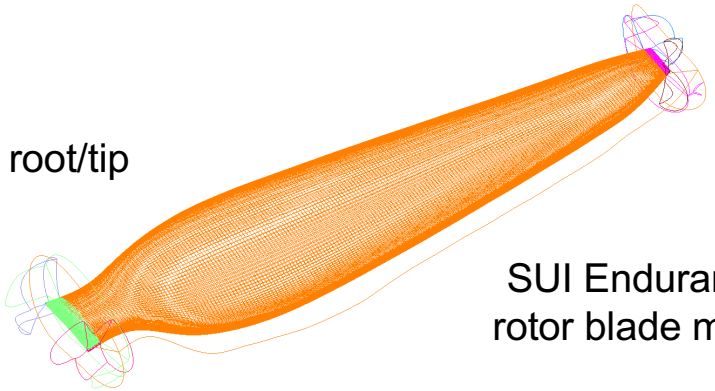
- Generate structured-overset rotor blade grids from conceptual design parameters

- Inputs
  - Airfoil coordinates (Selig format)
  - Spanwise definitions (taper, twist, sweep, droop)
  - Chord-wise & span-wise grid spacing control

- Outputs
  - Surface & volume meshes for rotor blade with flat root/tip



Tip grid topology [1]



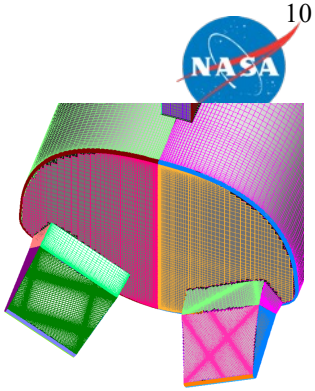
SUI Endurance rotor blade mesh

[1] Peters, N., and Shirzai, D., AIAA 2025-1068. *AIAA SCITECH 2025 Forum*. January 2025.

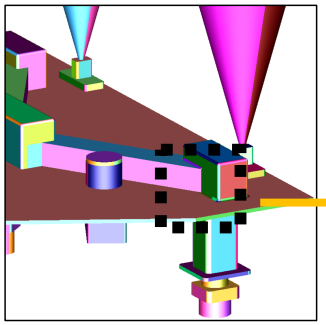
GEOMETRY PREPARATION FOR EPOGS

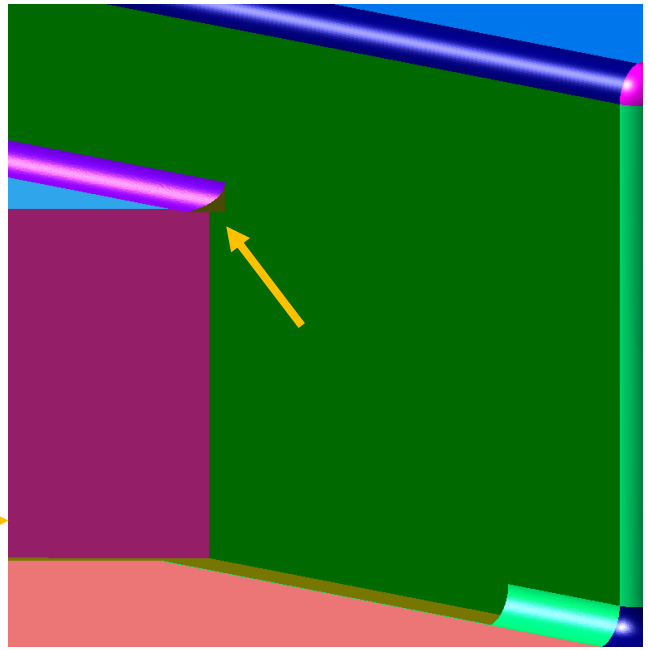
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TYPICAL GEOMETRY ISSUES

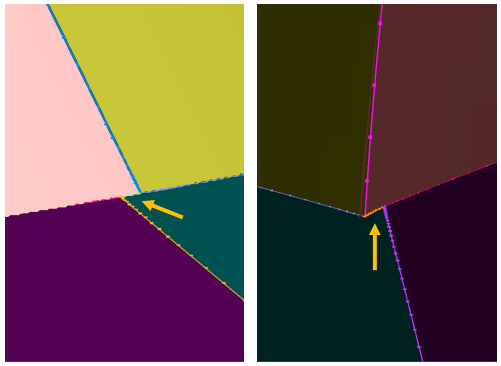


- CAD representation does not include original intent
  - Loss of left/right symmetry, body of revolution symmetry
- Translation loss from native CAD to neutral formats (.stp, .igs)
  - Intersection tolerances
  - Geometric components data
- Original geometry
  - Not closed (gaps, mismatches between surfaces)
  - Contains tiny irrelevant features for CFD simulations
  - Highly fractured
  - Highly trimmed

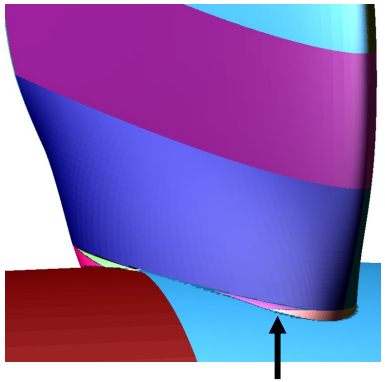




GEOMETRY BEST PRACTICES

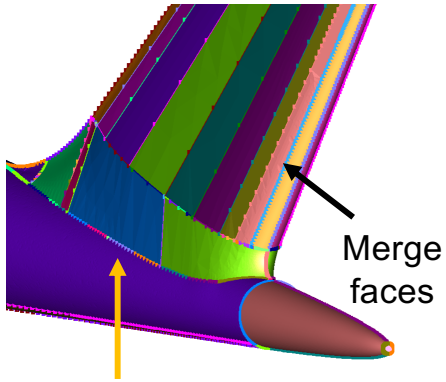


Remove tiny edges caused by tolerance settings



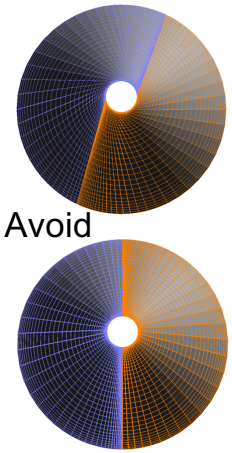
Remove sliver faces

- Remove spurious nodes
- Merge as many edges and faces as possible (total number of output grids ~ #faces + #edges + #nodes)
- Minimize use of trimming on the faces
- Merge tiny/sliver faces with adjacent faces
- Use symmetric faces/edges on symmetric geometry
- Use curvature-aligned face u/v parameters as much as possible
- Use hard edge to define leading edges of wings, flaps, slats, tails
- A face adjacent to the symmetry plane should have a hard edge along the symmetry plane
- No periodic faces (split into halves)



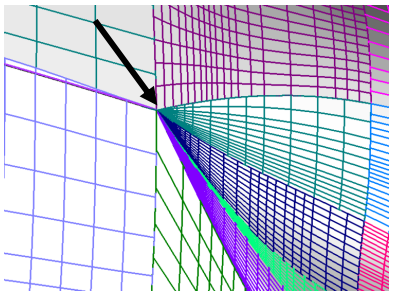
Merge faces

Remove spurious nodes

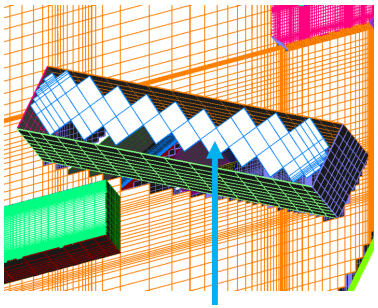


Avoid

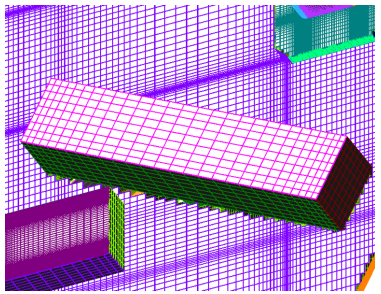
Do this



Avoid collapsed edge on smooth surface



Avoid excessive trimming



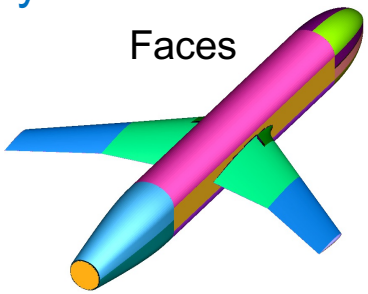
Aim for boundary-aligned u/v

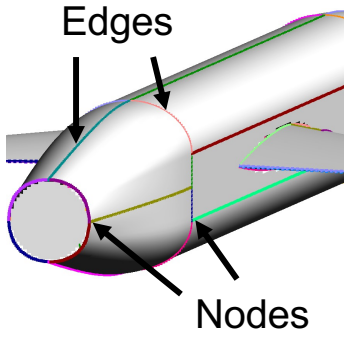
EPOGS AUTO-MESHING TOOL

BOUNDARY REPRESENTATION (BREP) GEOMETRY & MESHES

Input Geometry

- Boundary Representation (BRep) solid
- Interface: Engineering Sketch Pad (ESP) from MIT

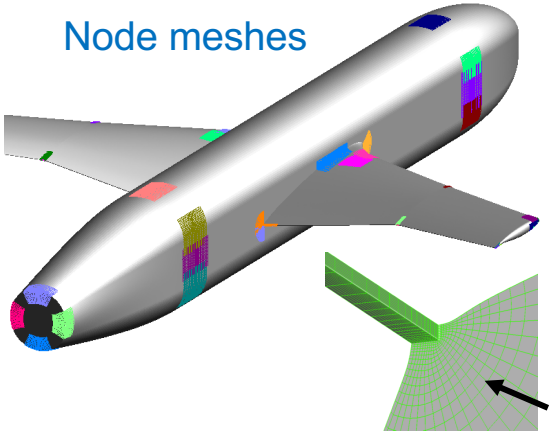




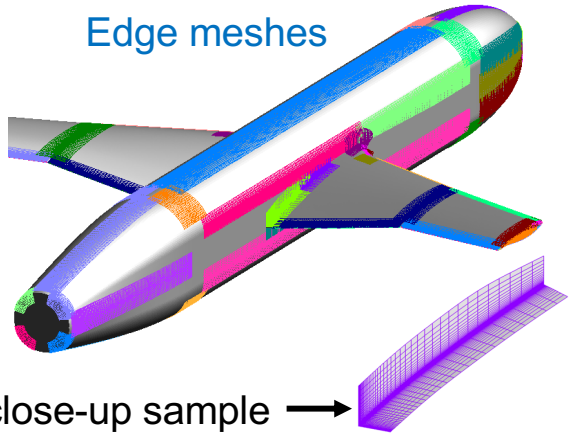
Auto-Generated Meshes

- More fractured than manually created meshes (Total # <= F + E + N)
- Does not appear to affect RANS flow solution aeroloads, convergence, cost (AIAA Paper 2023-3602)

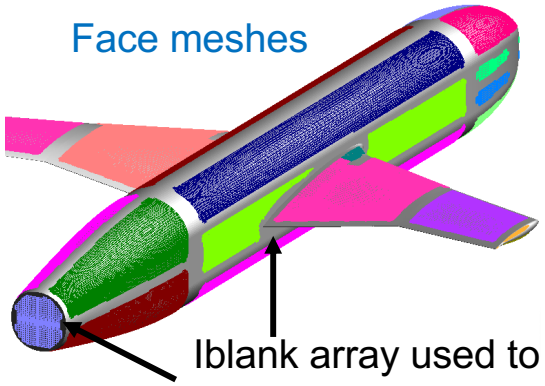
Node meshes



Edge meshes



Face meshes



TERMINOLOGY: CONFIGURATIONS, BODIES & COMPONENTS

Component

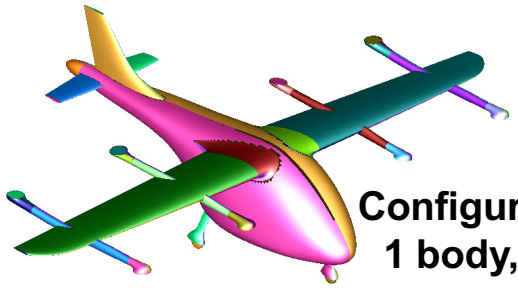
- A geometric part that can be separated from other parts by an intersection curve

Body

- Consists of one or more components that are connected
- Each component: multiplicity >= 1

Configuration

- Consists of one or more disjoint bodies
- Each body: multiplicity >= 1



Configuration with 1 body, multiple components

Configuration with 2 unique bodies & 16 total bodies

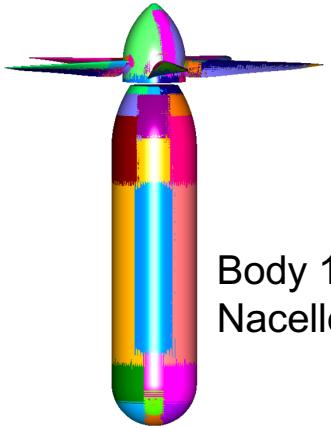
Body 1: Splitter plate, wing, nacelles, flap, brackets
Body 2: Blade x 15 (disjoint)



Configuration with 2 bodies

Body 2 (with 2 unique components, 6 total):

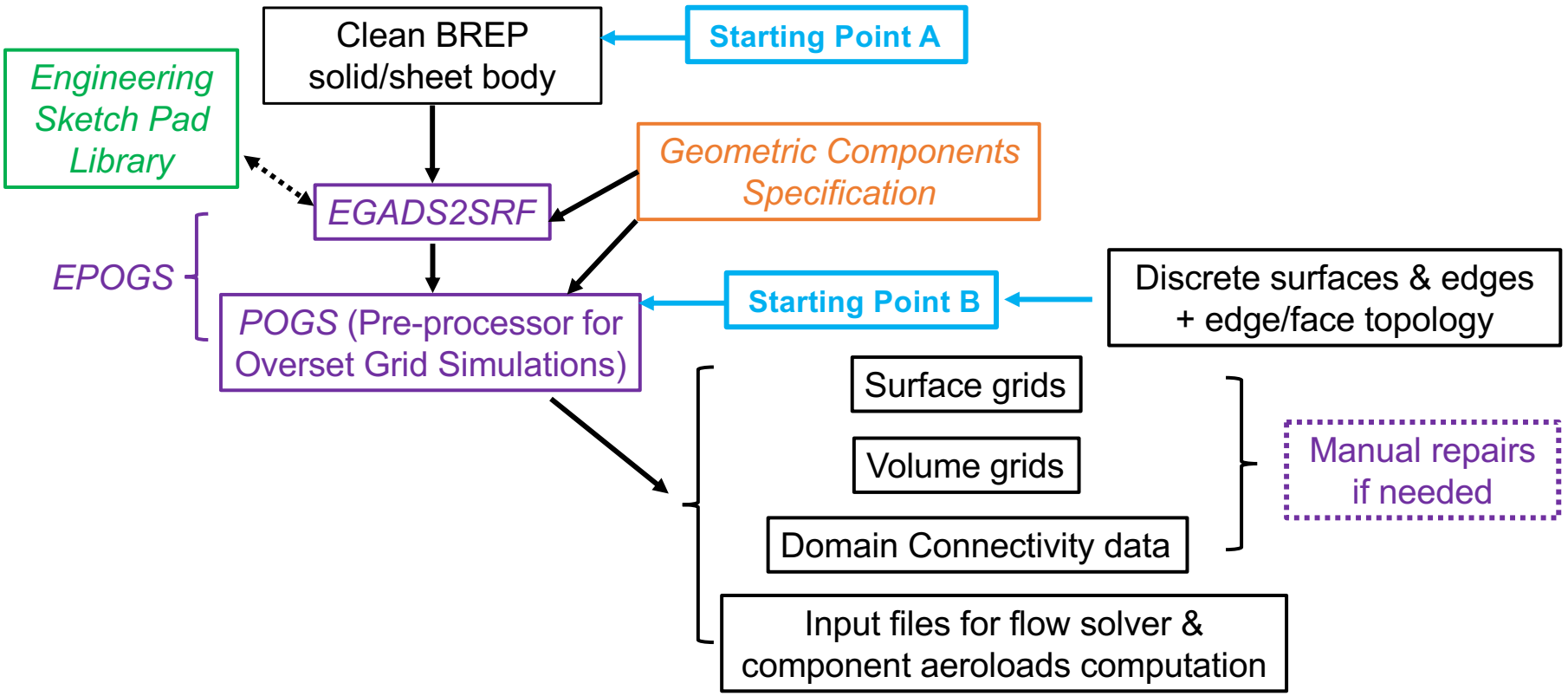
- Hub x 1
- Blade x 5



Body 1: Nacelle

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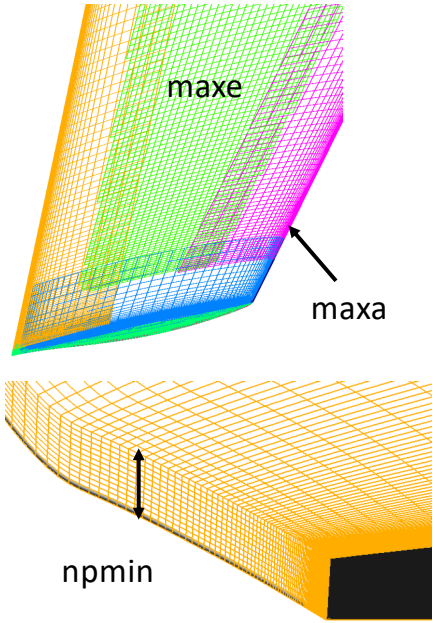
EPOGS FLOWCHART



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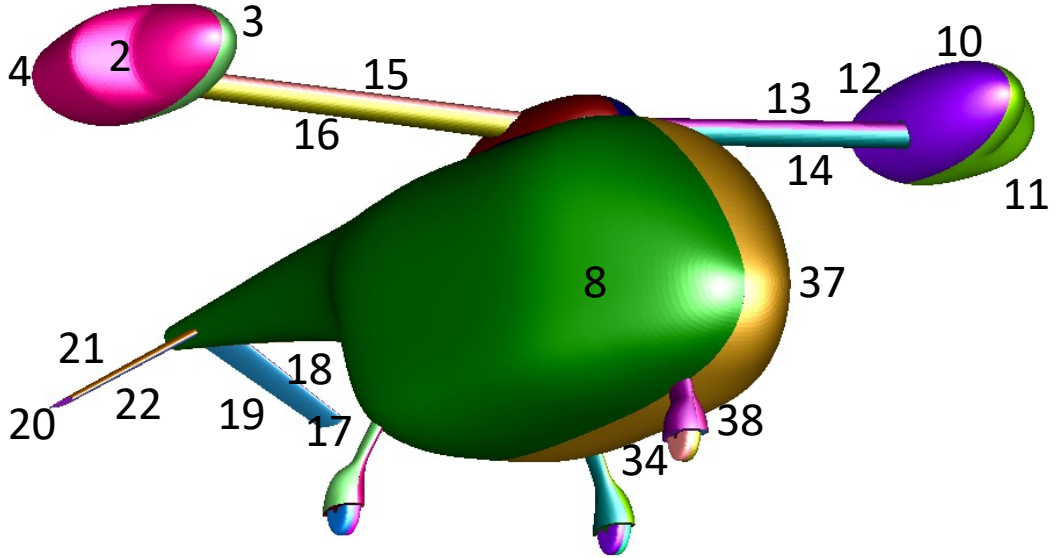
GLOBAL PARAMETERS

isym	0 = no symmetry 1/2/3/-1/-2/-3 = half-body in +x/+y/+z/-x/-y/-z 11/12/13 = full-body symmetric about x=0/y=0/z=0
refl	Reference length for local max grid spacing scaling
maxe	Max grid spacing (grid unit)
maxa	Max dihedral angle (degrees)
maxc	Max chordal deviation
maxsr	Max stretching ratio
npmin	Min # points on shortest edge (finite thickness t.e.)
dswall	Wall normal spacing in volume grids (grid unit)
nfringe	Number of fringe points layers at outer and hole boundaries
stenqual	Min acceptable stencil quality for orphan pts flagging
dobnd	Distance to far field boundary (grid unit or rel. near-body bbox)



GEOMETRIC COMPONENTS SPECIFICATION

- Text file: component name + list of face numbers
- Information usually not available from geometry file (STEP/IGES)
- Manual construction with OVERGRID + Text Editor
  (< 10 minutes for example below)
- Enables:
  - Local component characteristic length -> local max grid spacing
  - Auto construction of input file for component aero loads computation
  - Single component meshing option



Geometric Components Data File
(*geom*.001.gcomp)

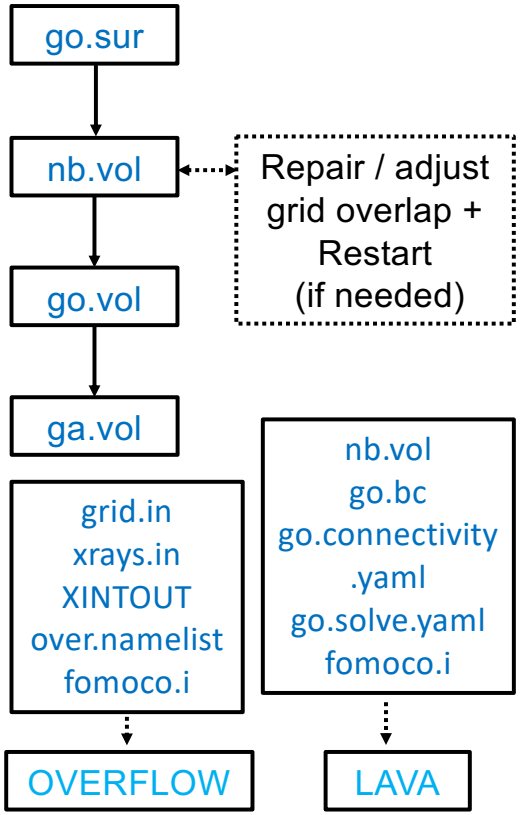
Right_LandGear 1,23:27	Left_Wing 13:14
Rotor_1_Cowling 2:4	Right_Wing 15:16
Main_Fuselage 5:6,8,37	Left_Tail 17:19
Fuselage_Fairing 7,9	Right_Tail 20:22
Rotor_2_Cowling 10:12	Left_LandGear 28:33
	Front_LandGear 34:36,38:40

EPOGS WORKFLOW & RESTART CHECK POINTS

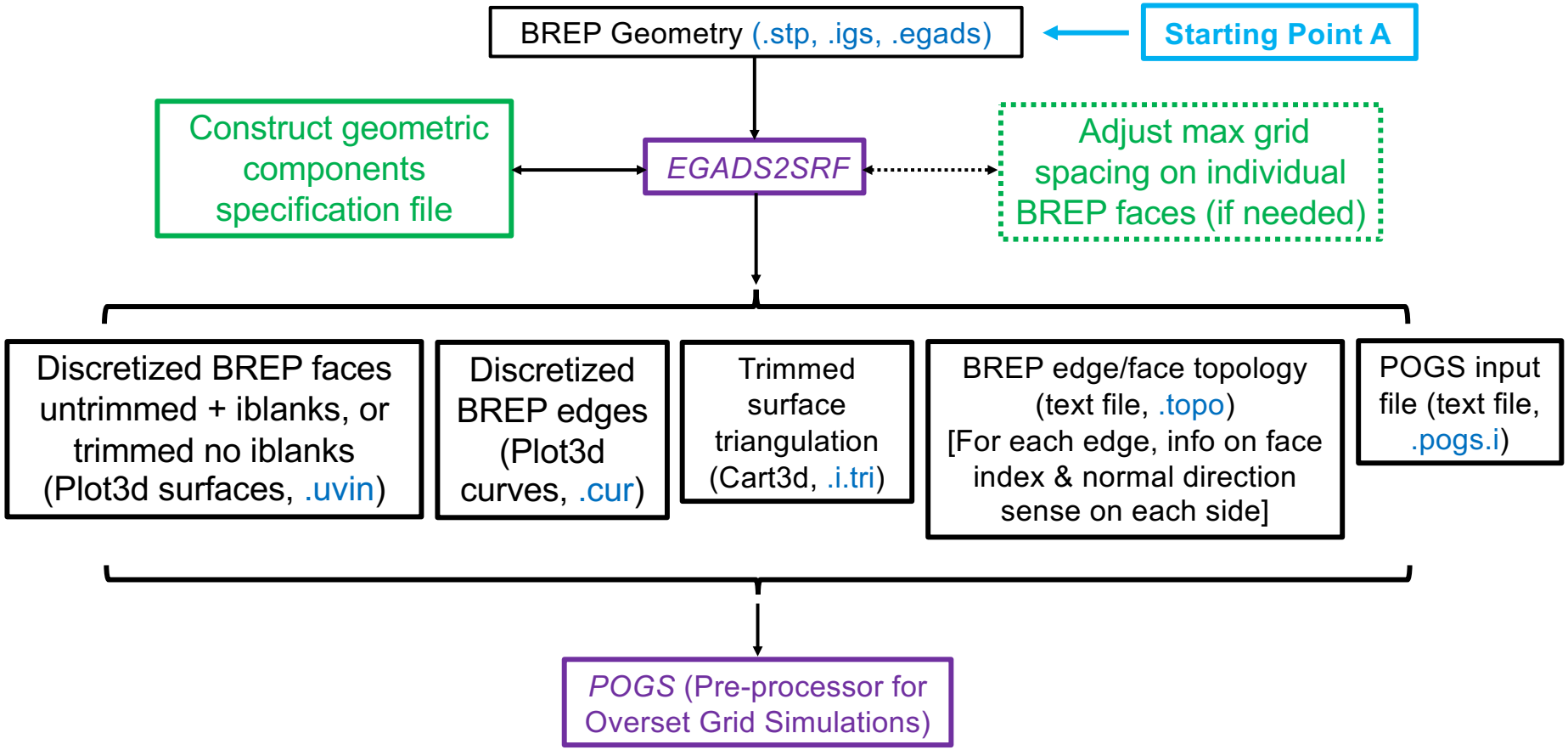
	Action/Command	Output File(s)	Output Type
1	*egads2srf -i root.stp -stop faces*	guvo.sur	Prelim discretized BREP faces
2	Construct components data file manually	root.001.gcomp	List of faces in each component
3	*egads2srf -i root.stp [args]*	root.001.{i.tri, uvin,cur,topo}	Discretized faces & edges, edge/face topology data
4	*pogs [args] -stop go.sur* < pogs.i	go.sur	Surface grids with connectivity data
5	*pogs [args] -start volgen -stop nb.vol* < pogs.i	nb.vol	Near-body volume grids
6	*pogs [args] -start boxgen -stop go.vol* < pogs.i	go.vol	Near & off-body volume grids
7	*pogs [args] -start vdocona -stop ga.vol* < pogs.i	ga.vol	Volume grids with domain connectivity data

root = rootname of geometry file

[args] = optional arguments



EGADS2SRF WORKFLOW



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EGADS2SRF

Command-Line Options

egads2srf -i ifile [-maxa maxang] [-maxe maxedge] [-maxc maxchord] [-isym sym] [-npmin np]
[-maxsr maxsrat] [-dswall dsw] [-nfringe nf] [–stenqual dqual] [-b bodygen] [-frm frmstring]
[-stop stoploc] [-c] [-r] [-help] **>! e.o**

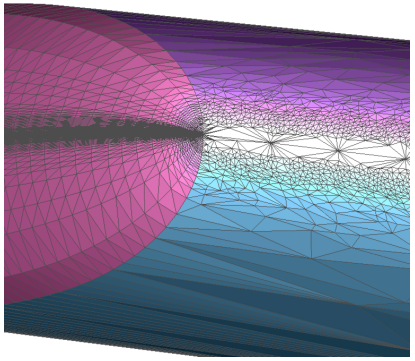
[ ] = optional inputs

ifile = input file containing geometry (igs, stp, egads)
maxang = max allowed dihedral angle in degrees (10.0)
maxedge = max allowed arc length of triangle edge or structured mesh spacing (-0.025)
> 0 take value as is, < 0 take absolute value times global bounding box diagonal arc length
maxchord = max allowed chordal deviation (-0.001)
> 0 take value as is, < 0 take absolute value times global bounding box diagonal arc length
sym = symmetry specification (0 = no symmetry)
= 1/2/3/-1/-2/-3 : geometry is a half body on the +x/+y/+z /-x/-y/-z side
= 11/12/13 : geometry is a full body symmetric about x=0/y=0/z=0
np = minimum number of points in each direction in u, v tessellation (9)
maxsrat = max stretching ratio for surface/volume grids (1.2)
dsw = wall normal spacing in volume grids
nf = number of fringe pt layers
dqual = minimum donor stencil quality
bodygen = solid body number to make grids on if more than 1 solid in ifile
frmstring = list of faces to remove from meshing domain for open geometry, e.g., 3,8,15:17
stoploc = stop location (contents/faces/ftable)

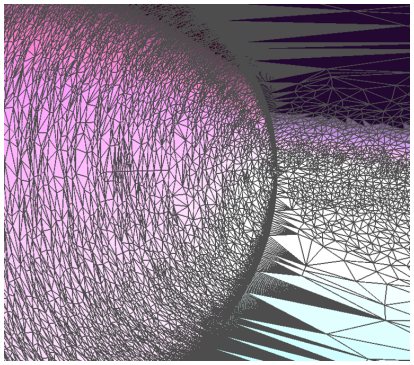
CHOICE OF MAX CHORDAL DEVIATION (maxc)

- Default value is sometimes too high
- Examine the triangulation for anomalies and reduce if necessary

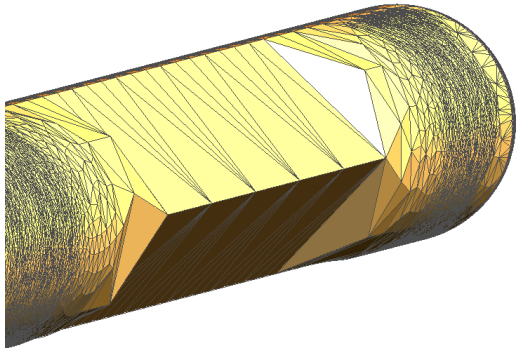
maxc = 0.05

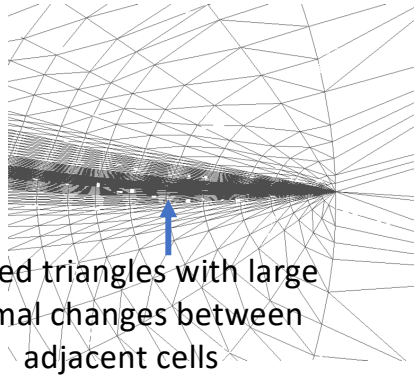


maxc = 0.002

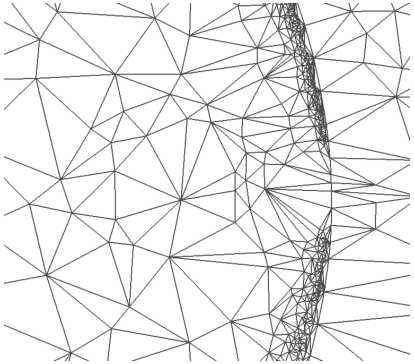


maxc = 0.1

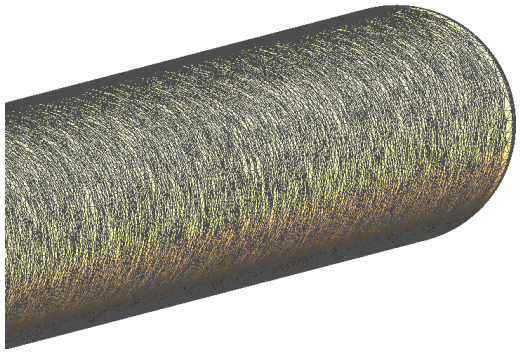




Skewed triangles with large normal changes between adjacent cells

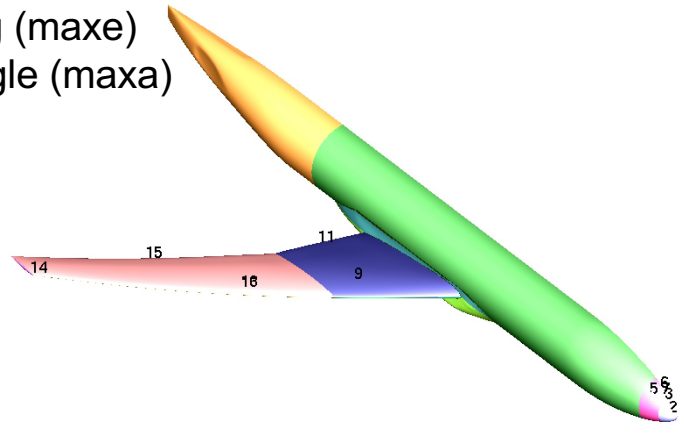


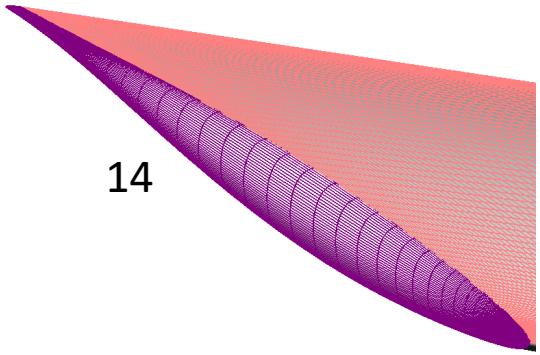
maxc = 0.01



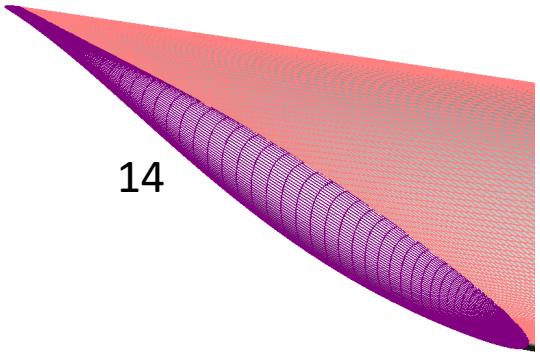
POINT DISTRIBUTION PARAMETERS FOR FACES

Editable text file
- max grid spacing (maxe)
- max dihedral angle (maxa)





maxe = 6.49163



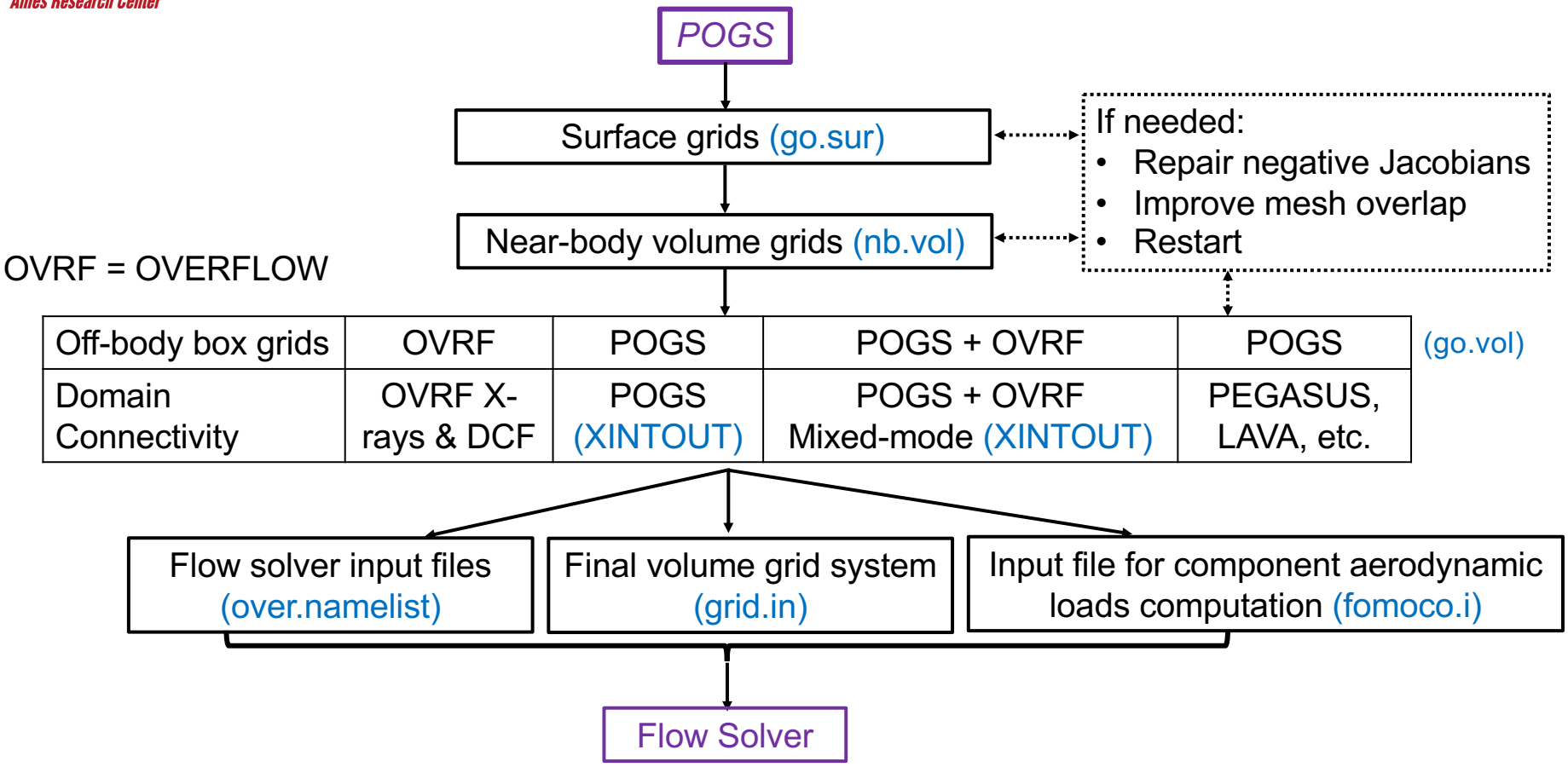
maxe = 3.25

Face Parameters File
(*geom*.001.face)

17	10.00000	4.00000	nface, maxe, maxa
! Face	max spacing	max angle	component
1	10.00000	4.10238	Fuselage
2	10.00000	4.10238	Fuselage
3	10.00000	4.10238	Fuselage
4	10.00000	4.10238	Fuselage
5	10.00000	4.10238	Fuselage
6	10.00000	4.10238	Fuselage
7	10.00000	4.10238	Fuselage
8	10.00000	4.10238	Fuselage
9	6.49163	4.00837	Wing
10	6.49163	4.00837	Wing
11	6.49163	4.00837	Wing
12	10.00000	4.10238	Fuselage
13	10.00000	4.10238	Fuselage
14	~~6.49163~~ 3.25	4.00837	Wing
15	6.49163	4.00837	Wing
16	6.49163	4.00837	Wing
17	6.49163	4.00837	Wing

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POGS WORKFLOW OVERVIEW



POGS: Command-Line Options

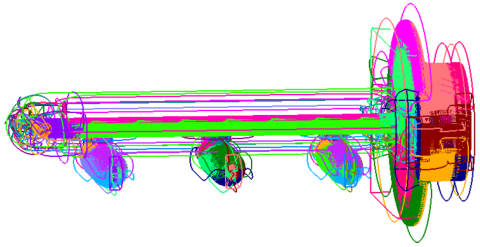
pogs [-nopro] [-arcse arc] [-refl reflen] [-nosplitg] [-dcfmix] [-dcfpogs] [-dcfxry]
[-combinebods nbody] [-splitsys] [-npmaxsp npsp]
[-start startloc] [-stop stoploc] **< *geom*.001.pogs.i >! pogs.o**

-nopro : begin with starting point B (BREP geometry not available)
arc: arc length for auto-activation of thin face logic (finite thickness t.e. arc length)
reflen: reference length for local component max grid spacing scaling (needed for -nopro)
-nosplitg: deactivate auto splitting of large single blocks that do not fit inside 32-bit record
-dcfmix : domain connectivity with OVERFLOW mixed-mode
-dcfpogs: domain connectivity with POGS
-dcfxry: domain connectivity with X-rays
-combinebods: number of bodies in relative motion (= #BREP solids)
- splitsys: split grid system with large single blocks (binary file) into system that fit 32-bit records
npsp: max number of grid points allowed in a single grid (use with -splitsys)

-start startloc : starting location
= volgen (make volume grids, needs go.sur, go.sub)
= boxgen (make off-body box grids after repair, needs go.sur, go.sub, go.vol)
= vdocona (volume connectivity part a, needs go,sur, go.vol, go.sub)
-stop : stoploc: stopping location
= go.sur (surface grids), go.vol (volume grids), ga.vol (volume connectivity)
= splitr.cur, thinf.sur, nm.{cur,sur}, em2.cur, enm.sur, gs.sur, tfnm.{.cur.sur} (for advanced users)

POGS WORKFLOW: SURFACE & VOLUME GRIDS

Discretized untrimmed faces with iblanks, Surface triangulation, Discretized edges, Edge/Face topology (generated by egads2srf or some other method)



POGS

Surface grids (go.sur)

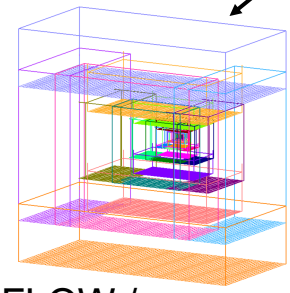
Near-body volume grids (nb.vol)

Starting Point B

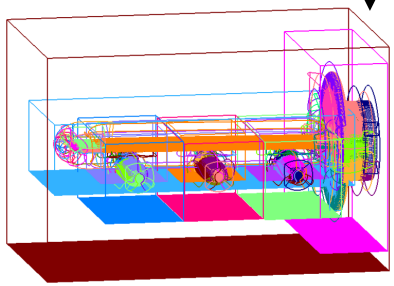
Repair & restart if needed:
- Negative Jacobians
- Improve mesh overlap

Off-body volume grid options

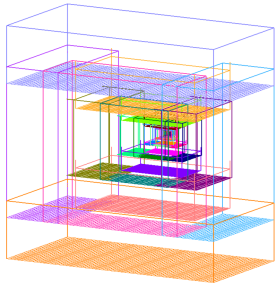
Flow solver	POGS	POGS + Flow solver



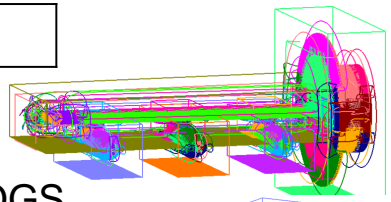
OVERFLOW / LAVA



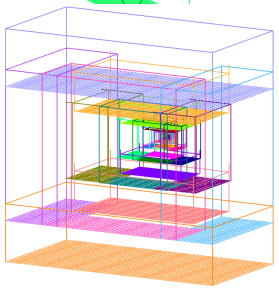
POGS



POGS



POGS



OVERFLOW / LAVA

POGS WORKFLOW: DOMAIN CONNECTIVITY

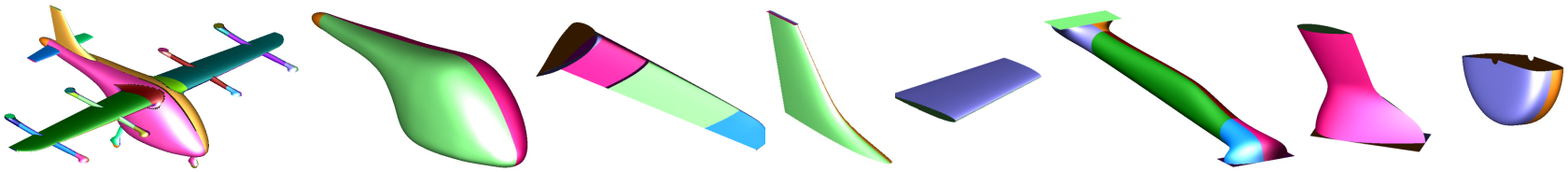
Near & Off-body volume grids (go.vol)

↓

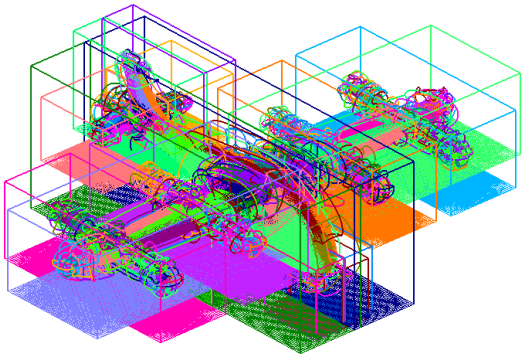
Options	OVERFLOW X-rays & DCF *	POGS (XINTOUT/ CINTOUT)	POGS + OVERFLOW Mixed-mode (XINTOUT/CINTOUT)	PEGASUS (XINTOUT) LAVA (INTOUT)

* POGS creates
- Component surfaces for X-ray map construction
- First guess at hole-cut instructions (in progress)

If needed:
- Improve mesh overlap
- Restart

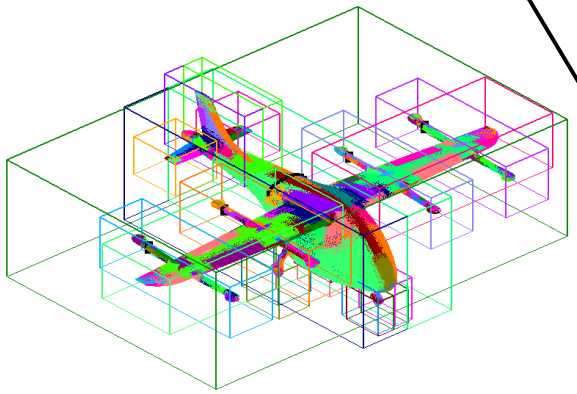


FILES NEEDED TO RUN OVERFLOW (MIXED-MODE)



Near-body curvilinear & component box volume grids
go.vol with enclosing outer box grid(s) removed
-> grid.in

POGS



X-ray maps for entire geometry & X-ray maps for component box grids
go.xry -> xrays.in

Forces/moments computation input file (default components defined in Geometric Components data file)
fomoco.i

OVERFLOW input file with auto-filled boundary conditions for each grid
over.namelist

Domain connectivity data (fringe points, donor stencils, hole-cut iblanks)
CINTOUT_NOGBOX -> CINTOUT

POGS: Input Files for Flow Solver & Force/Moment Computation Tool

Flow Solver

- OVERFLOW (over.namelist)
  - Boundary conditions for all grids
    - Viscous wall for all near-body grids
    - Symmetry plane (half-body cases)
    - Far field for off-body grid
  - Hole-cut instruction estimates (if using DCF)

- LAVA
  - Boundary conditions for all grids (go.bc)
  - Connectivity (go.connectivity.yaml)
  - Solver input (go.solve.yaml)

Force/moment computation tool FOMOCO/USURP (fomoco.i)

- Grid index ranges for each geometric component defined in *rootname*.001.gcomp

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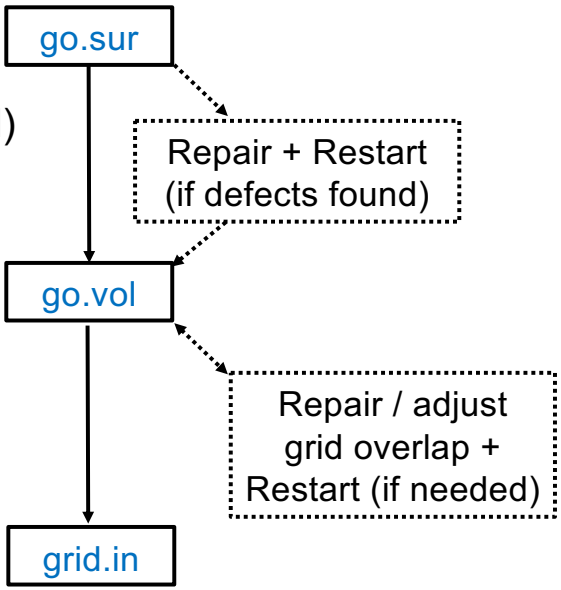
POGS Input & Output Files Summary

Input

- stp/igs/egads file with at least one solid (e.g., *geom*.stp)
- Geometric components specification file (e.g., *geom*.001.gcomp)
- Discrete surfaces/edges and topology (.uvin, .cur, .i.tri, .topo, .pogs.i)

Output:

1. Overset surface grids with iblanks: go.sur (Plot3d)
2. Overset volume grids without iblanks: nb.vol, go.vol (Plot3d)
3. Overset volume grids with volume connectivity iblanks: grid.in (Plot3d)
4. Domain connectivity data: XINTOUT/CINTOUT
5. X-ray map for entire geometry: go.xry -> xrays.in (X-ray)
6. Component forces/moments computation input file: fomoco.i (text)
7. Flow solver input files
   - OVERFLOW namelist input: over.namelist (text)
   - LAVA
     - boundary conditions go.bc
     - connectivity inputs go.connectivity.yaml
     - solver inputs go.solve.yaml



RUNEPOGS

(pronounced 'run– epogs')

A script called "runepogs" can be used to run a collection of cases

For each case, it performs the following:

- executes egads2srf followed by pogs

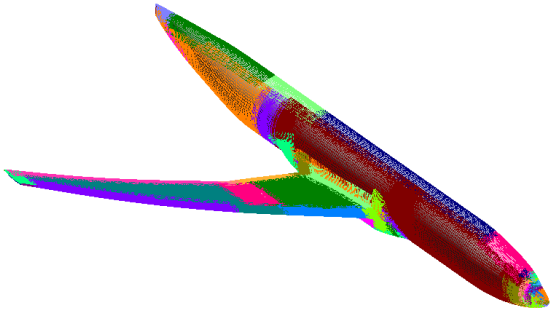
- reports
  - total number of volume grids
  - total number of volume grid points
  - number of defective surface and volume grids
  - wall clock time

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TUTORIALS & REPAIR SCRIPTS

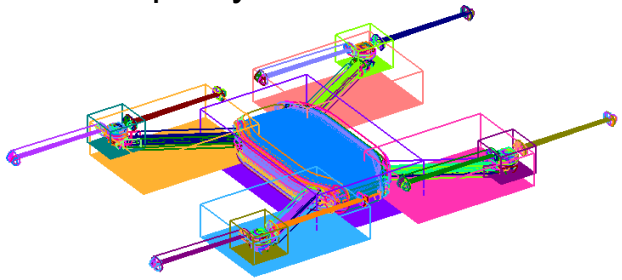
Wingbody

- Basic workflow & options



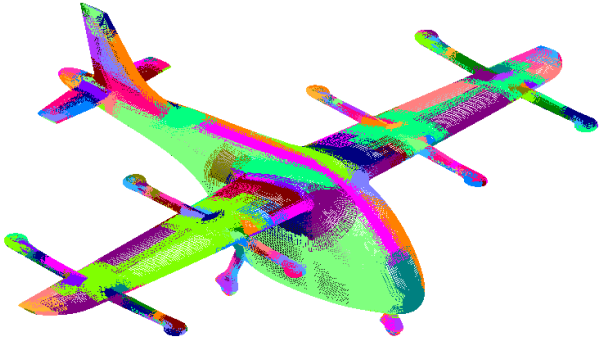
Drone

- Multiple disjoint bodies with multiplicity in relative motion



Lift+Cruise

- Symmetric body options
- Face mesh reconstruction



Grid Repair

- Negative Jacobians in near-body volume grids
  - Adjust hypgen smoothing/splay parameters
- Insufficient mesh overlap
  - Extrapolate in J/K on surface grid
  - Adjust splay in volume grid
- Example repair scripts in tutorials

RUNEPOGS

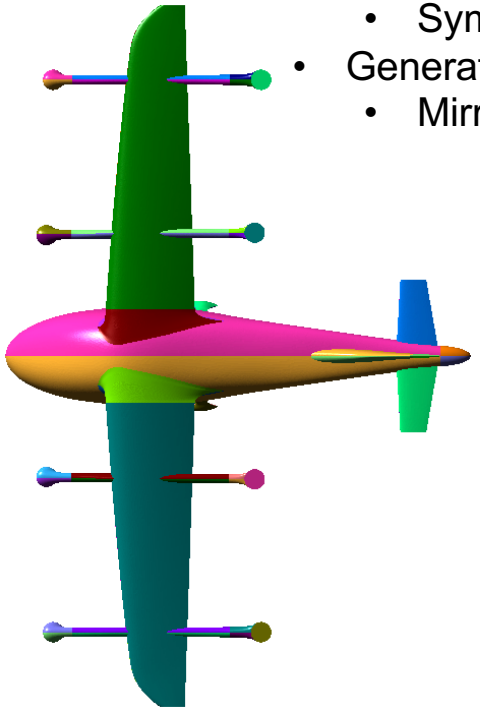
- Example on how to use the *runepogs* script to run a collection of cases

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CONFIGURATIONS WITH SYMMETRY OR GROUND PLANE

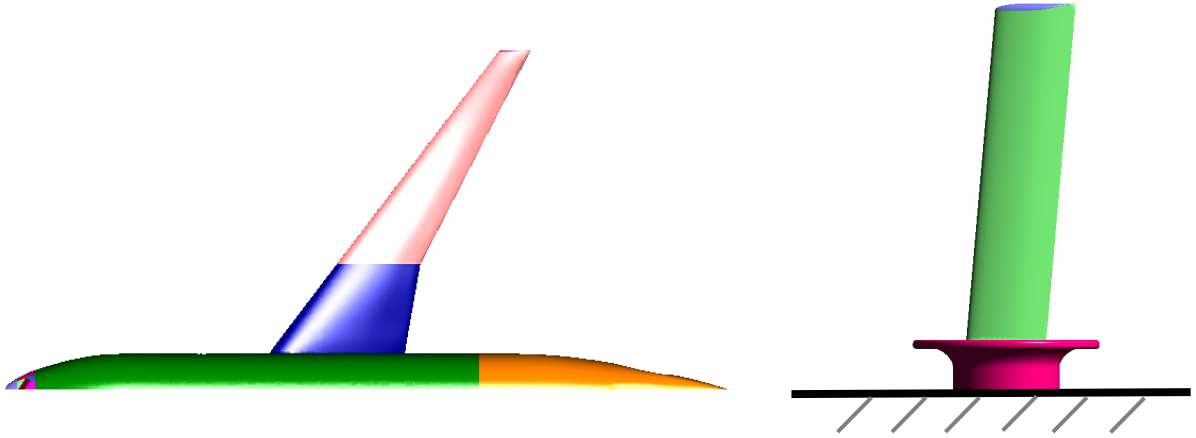
NASA

Input geometry: Full-body



- Generate grids on full body (-isym 0)
  - Symmetric grid system not guaranteed
- Generate grids on half-body (-isym 11/12/13 for x/y/z symmetry)
  - Mirror to guarantee symmetric grid system

Input geometry: Half-body or body on ground plane



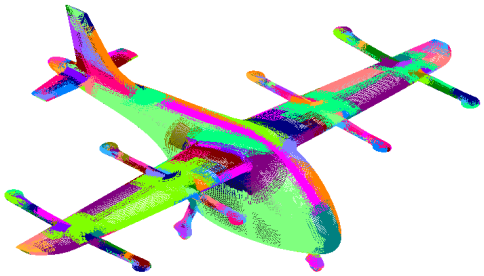
- Constant x/y/z plane on open boundary
  - -isym 1/2/3 for constant x/y/z

HOW TO HANDLE AUTOMATED MAX GRID SPACING SCALING IN A COMPONENT BUILD-UP STUDY *

Default behavior

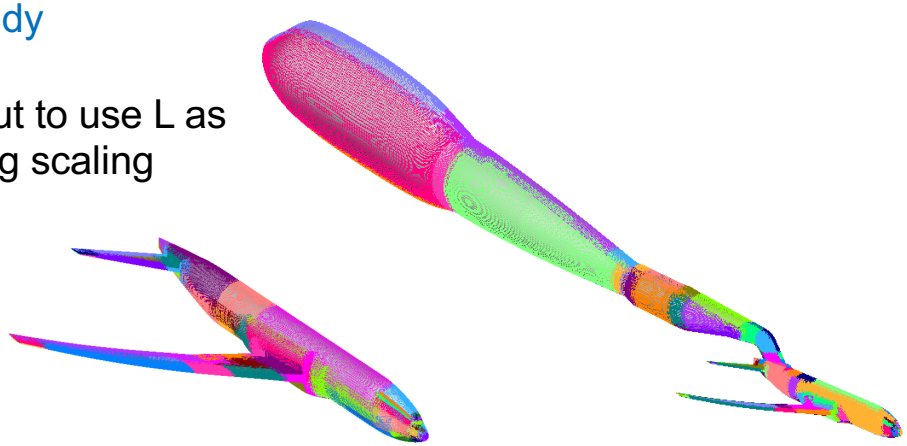
- Local max grid spacing for each component scaled by ratio of component bounding box max length to reference length L

  L = global bounding box max length



For consistent scaling in component build-up study

- Specify “-refl L” in EPOGS command line input to use L as the reference length for local max grid spacing scaling



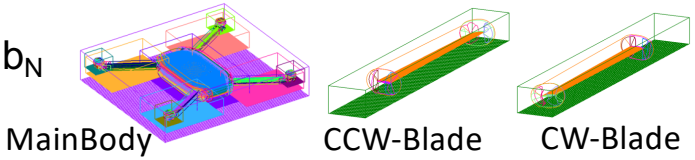
* CGT 3.0+

HOW TO HANDLE GRID SYSTEM WITH VERY LARGE SINGLE-BLOCKS IN ONE OR MORE GRIDS

- Run pogs with “-nosplitg”
  - Resulting grid files (nb.vol, go.vol, ga.vol) in binary format (no record markers)
- Run pogs with “-splitsys -npmaxsp *npsp*”
  - Convert volume grid files (nb.vol, go.vol, ga.vol) into Fortran unformatted format
  - Split grids that are larger than *npsp* points into two or more blocks
  - Adjust flow solver inputs and force/moment computation inputs appropriately

HOW TO HANDLE CONFIGURATIONS WITH MULTIPLE DISJOINT BODIES WITH MULTIPLICITY

- In main configuration directory, create subdirectories b_1, b_2, … b_N
  N = number of disjoint bodies



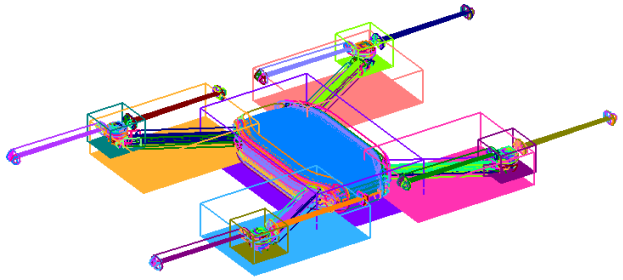
- Run EPOGS for each body in its own directory to generate all surface & volume grids, domain connectivity data, X-ray map, flow solver input, fomoco input
  (go.sur, nb.vol, go.vol, ga.vol, XINTOUT, go.xry, over.namelist, fomoco.i)

- In main configuration directory, run pogs with

  "-combinebods" N –bodymult bmults –bodyname bnames"

  bmults = multiplicity for each body (e.g., 1,4,4)

  bnames = name for each body (e.g., MainBody, CCW-Blade, CW-Blade)



to generate merged grid, domain connectivity, X-ray map, flow solver and fomoco inputs

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IN PROGRESS: HOW TO HANDLE
(1) DUPLICATED COMPONENTS IN THE SAME BODY
(2) COMPONENT BUILD-UP STUDY

Use geometric components data file to specify

- Multiplicity of duplicated components in same body
- Components in build-up study

Original format

```
Hub
1:12

Blade
13:18
```

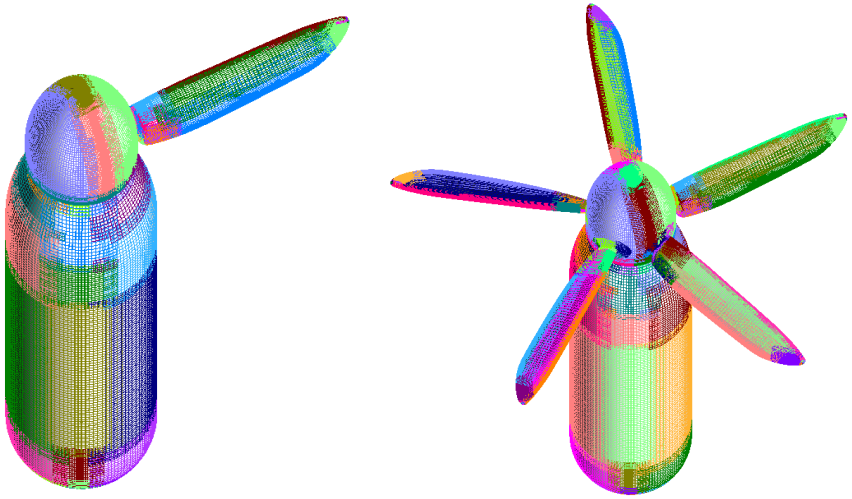
Extended format

```
Hub     1
1:12

Blade   5
13:18

Mainbody   0
19:23
```

MULT >= 1 component multiplicity
= 0 exclude component

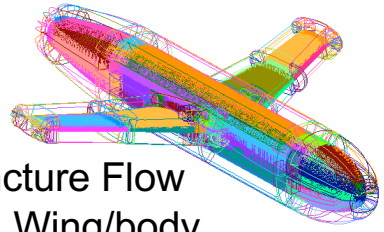


- Generate grid system with multiplicity
- Generate X-ray map with multiplicity
- Generate Config.xml file with grid list for each component
- Update flow solver and fomoco inputs

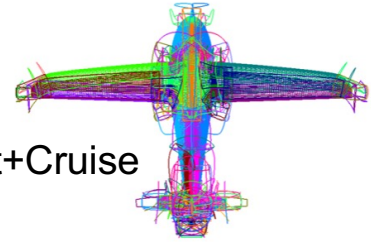
CONCERNED ABOUT FRACTURED EPOGS GRIDS COMPARED WITH MANUAL MESHING APPROACH?

Flow solver - OVERFLOW

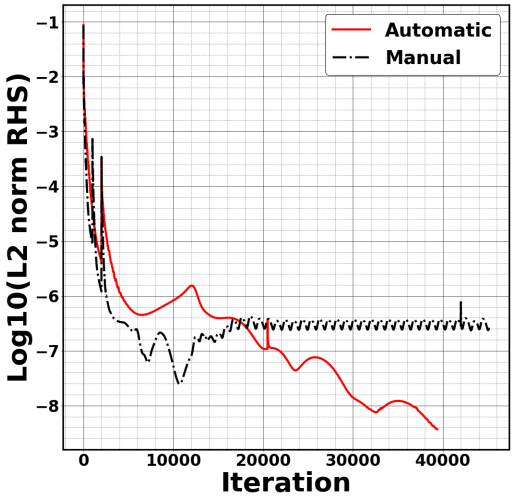
AIAA Paper 2023-3602

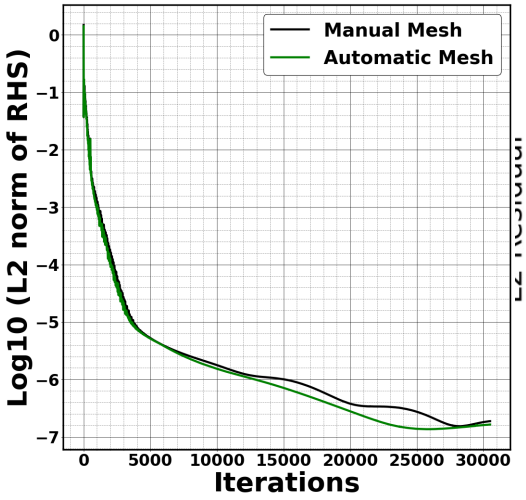


Juncture Flow
Exp. Wing/body



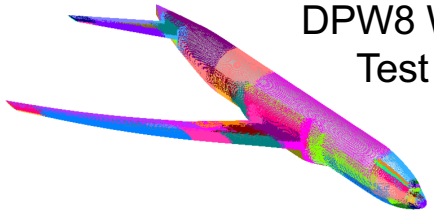
Lift+Cruise



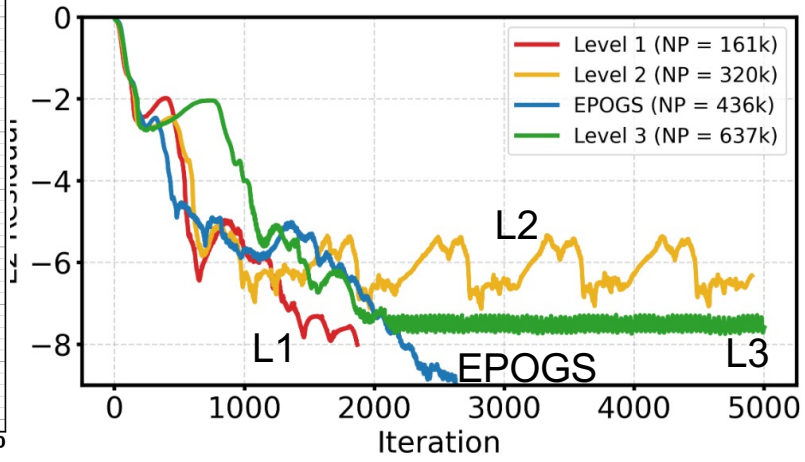


Flow solver - LAVA

Abstract in preparation for Aviation 2027



DPW8 Wing/body/tail
Test Env Study



RECENT PUBLICATIONS

Chan, W. M., Chuen, A. M., Jensen, J. C., Shauerhamer, D. G., Overset Structured Mesh Generation on Complex Configurations Using the EPOGS Toolset, AIAA Paper 2026-4121, June, 2026.

Jensen, J. C., Chuen, A. M., Chan, W. M., Pre-Test CFD Analyses for the Tiltwing Test Rig Experiment, AIAA Paper 2025-3384, July, 2025.

Chan, W. M., Chuen, A. M., Advances in Automation of Overset Structured Volume Mesh Generation and Domain Connectivity, AIAA Paper 2024-4305, Aug, 2024.

Hosseini, S. S., Chuen, A. M., Chan, W. M., Computational Aerodynamic Study of the Lift+Cruise VTOL Concept Vehicle Components, Transformative Vertical Flight 2024, Feb 2024.

Chuen, A. M., Hosseini, S. S., Jensen, J. C., Chan, W. M., Aerodynamic Simulations for Complex Geometries Using Automatically Generated Structured Overset Meshes, AIAA Paper 2023-3602, June, 2023.

Chan, W. M., Pandya, S. A., Chuen, A. M., Automation of Overset Structured Surface Mesh Generation on Boundary Representation Geometries, AIAA Paper 2022-3607, June, 2022.

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PRESENTATIONS

Chan, W. M., Chuen, A. M., Jensen, J,. C., Recent Development and Applications of the EPOGS Toolset for Automatic Structured Overset Mesh Generation, *AMS Seminar Series*, Jan., 2025.

Chuen, A. M., Hosseini, S. S., Jensen, J. C., Chan, W. M., A Study on Flow Solutions Computed on Automatically and Manually Generated Structured Overset Meshes, *AMS Seminar Series*, Sept., 2023.

Chan, W. M., Chuen, A. M., Recent Developments in Automation of Overset Structured Mesh Generation, *AMS Seminar Series*, Jan., 2022.

ACKNOWLEDGEMENTS

Funding from NASA Projects

Subsonic Vehicle Technologies (SVT)
Integrated Methods and Research (IMR)
Subsonic Flight Demonstrator (SFD)

Computational Resources

NASA Advanced Supercomputing (NAS) facility

DEMO

- How to construct the geometric components data file *(.gcomp)*
  - Using OVERGRID & editor
  - Using Rhino plug-in
- Analytic geometry starting point (A)
  - Wingbody test case (simple case to show workflow and basic options)
  - Drone test case (multiple disjoint bodies with multiplicity & relative motion dynamics)
- Discrete geometry starting point (B)
  - Podwing test case (input files from OpenVSP)

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GEOMETRIC COMPONENTS SPECIFICATION

